

# RECYCLING EV BATTERIES: An Elephant In The Room?

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**E**lectric vehicle (EV) is a hot topic in the automotive world. India's push towards commercialising EVs on a large scale within the next decade has set the stage for shaping clean and green roads of the future. But has India thought out all possibilities and challenges it might face going ahead?

One major component of EVs is the battery. While Indian EV battery providers are slowly shifting from lead-acid batteries to lithium-ion ones, ideas for recycling these need to be discussed right away. In this article we look at the importance of battery recycling and how far India has thought about it.

## The necessity of battery recycling

As majority of the world has shifted its preference to lithium batteries, demand for lithium is increasing exponentially. Since it is not an abundant element on Earth, it is necessary to reuse lithium from used batteries in a proper way to keep up with the supply.

Moreover, improper disposal of these batteries is a critical source of environmental pollution, and is capable of poisoning water

bodies, soil and air. Lithium-ion batteries are highly reactive and carry a high risk of explosion. On the other hand, lead-acid batteries, which are also significantly used in India, can lead to lead poisoning, which creates severe health risks.

International records show that in the European Union (EU), only five per cent of the total lithium-ion batteries—available across all kinds of products ranging from EVs to consumer tech—gets recycled. The remaining are either not disposed at all or are simply thrown away.

On the other hand, at a global scale, it is reported that, about 11 million lithium batteries are projected to get accumulated by 2030, considering the pace at which EVs are becoming popular. India is also a part of these predictions and, hence, requires a strong battery-recycling policy and infrastructure in order to be future-ready.

## The world addressing recycling requirements

Experts from Nissan and Mahindra Electric, among others, believe that a lithium-ion battery for EVs, after serving its tenure (up to 10 years), still preserves 70 per cent of its capacity. These can be tested and repackaged into more static energy-storage applications, like in homes, offices, streetlights, data centres and so on. The batteries can run for another five years in these applications. This process is being applied internationally (mainly in Europe, China and to some extent in the US). It is an effective method that Indian battery solution providers can apply to mitigate battery-disposal issues here.

This step, however, does not address the issue of sourcing elements like lithium and cobalt, which are essential to produce new EV batteries. Globally, utilising older batteries to extract sufficient lithium is under R&D, as solution providers are trying to find the most cost-effective and optimal method with the least amount of risk.

